



Mario Padilla Franco

Software Engineer focused on AI & Robotics

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Professional Profile

Engineer with professional experience across software, artificial intelligence, robotics and industrial automation. Built, integrated and supported production systems ranging from conversational AI platforms to robotic cells and industrial automation, with hands-on experience in Python, C++, Linux, TypeScript/JavaScript, QA, commissioning and multidisciplinary troubleshooting.

Professional Experience

AI Engineer

HubTalk

Feb 2026 - Present

- Design and maintain production voice and chat AI agents with prompt architecture, node-based flows, dynamic variables, guardrails, routing and escalation logic.
- Run structured QA using test cases, edge cases, stress scenarios, regression checks, call reviews and failure-mode analysis; use Cekura for test evidence.
- Built an internal Next.js/React agent tester for configuring dynamic variables, making web calls and exporting QA evidence.
- Support voice infrastructure troubleshooting across SIP/VoIP providers, FusionPBX, logs and platform configuration.

Application Engineer

GUREGO

May 2023 - Jan 2026

- Integrated and commissioned collaborative robots, AMRs, PLC-controlled systems, industrial vision solutions and production cells.
- Programmed and commissioned 10+ industrial and collaborative robots for palletizing, welding, pick-and-place, screwdriving, polishing and quality inspection.
- Troubleshoot robot motion, PLC signals and I/O, sensors, tooling, vision, safety devices, application logic, Python/C++ code, Linux and networking.
- Provided remote and on-site support, technical documentation and production incident resolution for customer deployments.

Assembly Engineer

Alliance Automation

Apr 2022 - Mar 2023

- Built custom automation equipment including conveyors, palletizers, sorting systems, robotic subsystems and motion-control assemblies from engineering documentation.
- Installed and tested sensors, actuators, control panels and safety devices; resolved mechanical, wiring and integration issues before commissioning.

Automation Engineer

Parque Tecnológico Iberoingenio

Aug 2021 - Apr 2022

- Supported deployment of an Industry 4.0 laboratory integrating robots, PLCs, remote I/O, sensors and safety systems.
- Prepared technical documentation and supported testing and demonstrations of advanced manufacturing technologies at Hannover Messe Leon 2021.

Education

2026 - Present

Master in Computer Science and Technology

Universidad Carlos III de Madrid

Current postgraduate studies in advanced computer science and computing technologies (60 ECTS).

2024 - 2025

Professional Master's Degree in Robotics

TECH Universidad

University-specific degree (Titulo Propio), 60 credits and 1,500 hours.

2017 - 2021

Bachelor's Degree in Mechatronics Engineering

Universidad Iberoamericana

Graduated with honors for academic excellence.

Certifications

IBM Full Stack Software Developer Professional Certificate · IBM, In progress

DevOps Basics for Everyone · IBM, May 2025

EF SET English Certificate - C1 Advanced · EF SET, 2024

Crash Course on Python · Google, 2024

Languages

Spanish: Native · English: C1 Advanced
- EF SET certified

Selected Projects

Industrial Palletizing Automation

[Doosan](#) · [Robotics](#) · [Palletizing](#) · [Automation](#)

- Developed configurable pallet patterns, frame-based pick/place positions and dynamic approach/retract poses for a Doosan robot cell.
- Implemented vacuum/tool-state handling, recovery sequences, continuity counters and motion safeguards for production reliability.

Conversational Agent Engineering (CAE) Framework

[OpenAI API](#) · [Cekura](#) · [Retell](#) · [LiveKit](#) · [LLM Agents](#) · [Voice AI](#)

- Structured business requirements into reusable conversation architectures, prompt patterns, routing rules and variable models.
- Defined QA processes, quality metrics and failure analysis covering edge cases, adversarial scenarios and regression testing.

Study Copilot - Lecture Transcription & Notes Pipeline

[Next.js](#) · [TypeScript](#) · [Python](#) · [Supabase](#) · [Whisper](#) · [FFmpeg](#) · [OpenAI API](#)

- Built a Next.js/TypeScript interface and Python workers for audio/PDF processing, Whisper transcription and structured note generation.
- Used Supabase for queue and state management, private result storage, job claiming and failure handling.

Industrial Robot Programming & Commissioning

[Robotics](#) · [Commissioning](#) · [Welding](#) · [Pick-and-Place](#) · [Screwdriving](#) · [Polishing](#)

- Programmed and commissioned 10+ industrial and collaborative robots for welding, pick-and-place, screwdriving, polishing, palletizing and quality inspection.
- Performed robot configuration, tool and frame setup, I/O integration, motion validation, start-up and troubleshooting for automotive and consumer-goods customers.

Doosan ROS 2 Motion Planning Lab

[ROS 2 Jazzy](#) · [Python](#) · [C++](#) · [Docker](#) · [MoveIt 2](#) · [RViz 2](#) · [Gazebo](#) · [Doosan](#)

- Built a Docker-based ROS 2 Jazzy simulation environment for a Doosan M1013 robot on Ubuntu 24.04.
- Worked with modular motion execution, robot abstraction and planning/visualization using MoveIt 2, RViz 2 and Gazebo.

Technical Skills

Software Development

[Python](#) · [C++](#) · [JavaScript](#) · [TypeScript](#) · [HTML](#) · [CSS](#) · [React](#) · [Next.js](#) · [Node.js](#) · [Express](#) · [SQL](#) · [Git](#) · [GitHub](#) · [Linux](#) · [Docker](#)

Robotics & Simulation

[ROS 2](#) · [MoveIt 2](#) · [RViz 2](#) · [Gazebo](#) · [Webots](#) · [RoboDK](#) · [Doosan](#) · [Universal Robots](#) · [DOBOT](#) · [MiR AMRs](#)

Industrial Vision

[Solomon](#) · [Mech-Mind](#) · [SICK](#) · [Cognex](#) · [Vision-Guided Automation](#) · [Quality Inspection](#)

AI & Voice Systems

[OpenAI API](#) · [Whisper](#) · [Cekura](#) · [Retell](#) · [LiveKit](#) · [LLM Agents](#) · [Voice Agents](#) · [Prompt Engineering](#) · [Conversation Architecture](#) · [Dynamic Variables](#) · [Guardrails](#) · [Routing & Escalation](#) · [Structured QA](#) · [Regression Testing](#) · [Edge-Case Analysis](#) · [Failure-Mode Analysis](#)

Automation & Controls

[PLCs](#) · [Siemens](#) · [Allen Bradley](#) · [Mitsubishi](#) · [PLC Signals](#) · [I/O Mapping](#) · [Sensors](#) · [Actuators](#) · [Safety Devices](#) · [Safety Scanners](#) · [Control Panels](#) · [Calibration](#) · [Production Troubleshooting](#)

Networking & Voice Infrastructure

[SIP/VoIP Troubleshooting](#) · [FusionPBX](#) · [Network Diagnostics](#) · [Linux Networking](#) · [TCP/IP Fundamentals](#)